Student Number:…………………………

2022 Higher School Certificate
Trial Examination
Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- Write your student number on the paper and the multiple-choice answer sheet
- A reference sheet is provided at the back of this paper
- In Questions 11-35, show relevant mathematical reasoning and/or calculations
- Diagrams not to scale

Total marks:
100

Section I
10 marks
- Attempt Questions 1-10 (pages 2-6)
- Allow about 15 minutes for this section

Section II
90 marks
- Attempt Questions 11-35 (pages 7-30)
- Allow about 2 hours and 45 minutes for this section

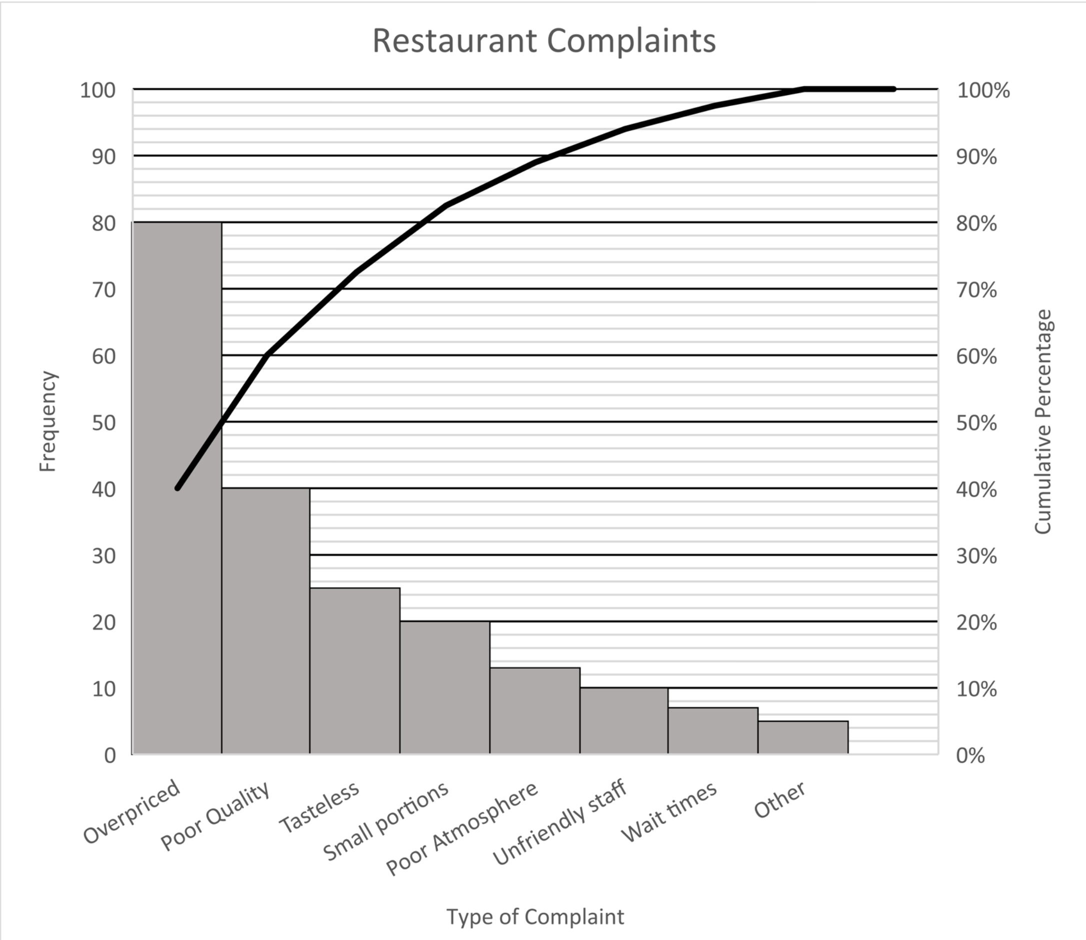
MATR22_EXAM

Section I

10 marks
Attempt Questions 1-10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-10

1 The pareto chart below shows the frequency of types of complaints received by a restaurant.



What percentage of complaints were for poor quality?

A. 20%

B. 40%

C. 60%

D. 80%

2 A letter is chosen at random from the word CALCULATOR.

What is the probability that the letter is a vowel?

A. 0.2

B. 0.4

C. 0.6

D. 0.8

3 The mean of a set of five numbers is 11. A sixth number is added to the set and the mean decreases to 10.

What is the sixth number?

A. 1

B. 3

C. 5

D. 7

4 What is the exact value of $\sin 135° + \tan 300°$?

A. $\frac{\sqrt{2}+2\sqrt{3}}{2}$

B. $\frac{\sqrt{2}-2\sqrt{3}}{2}$

C. $\frac{\sqrt{5}}{2}$

D. $\frac{-\sqrt{5}}{2}$

5 Find: $\int \frac{x}{x^2+3}\,dx$

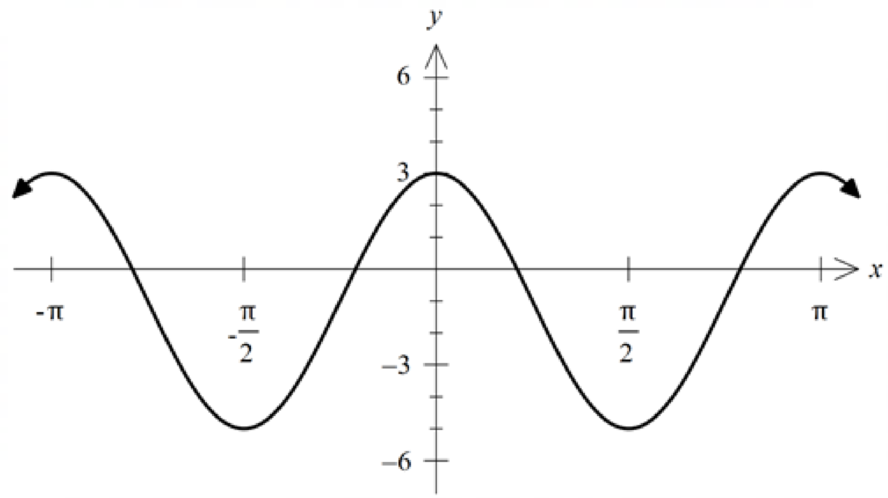
A. $\ln|x^2 + 3| + C$

B. $2\ln|x^2 + 3| + C$

C. $\frac{1}{2}\ln|x^2 + 3| + C$

D. $-2\ln|x^2 + 3| + C$

6 The graph of a trigonometric function is shown below.



What is a possible equation of the trigonometric graph shown?

A. $y = 3\cos x - 1$

B. $y = 3\cos 2x - 1$

C. $y = 4\cos x - 1$

D. $y = 4\cos 2x - 1$

7 The table shows the probability distribution for rolling a biased die.

X	1	2	3	4	5	6
$P(X)$	0.35	0.3	0.2	β	0.05	0.05

What is the value of β?

A. 0.05

B. 0.1

C. 0.15

D. 1

8 The graph of $y = g(x)$ has been transformed by translating it vertically by 3 units and then dilating it horizontally by a factor of 2.

Which of the following is the equation of the transformed graph?

A. $y = 2g(x) + 3$

B. $y = 2g(x + 3)$

C. $y = g(2x) + 3$

D. $y = g\left(\frac{x}{2}\right) + 3$

9 What is the gradient of the tangent to the function $y = \cos^2 x$ at $x = \frac{\pi}{6}$?

A. -1

B. $-\frac{\sqrt{3}}{2}$

C. $\sqrt{3}$

D. $\frac{1}{2}$

10 It is known that $f(x)$ is an odd function and $g(x)$ is an even function.

Given that $f(2) = 2$ and $g(2) = -2$, what is the value of $f\big(g(-2)\big) + g\big(f(-2)\big)$?

A. -4

B. -2

C. 0

D. 4

Section II

90 marks
Attempt Questions 11-35
Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

In Questions 11-35 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (1 mark)

Differentiate $(x + 4)^7$ **1**

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Question 12 (2 marks)

Calculate the exact value of **2**

$$\int_0^1 e^{5x}\, dx$$

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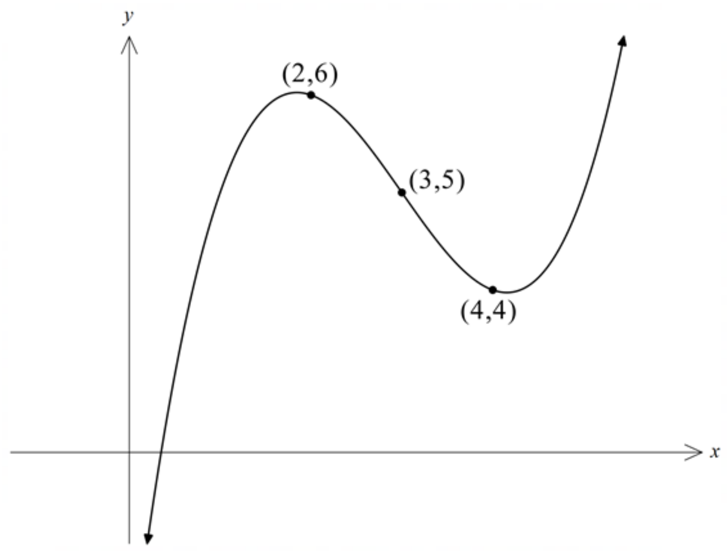
Question 13 (3 marks)

The graph of $y = f(x)$ is shown. **3**



Use two applications of the trapezoidal rule to find an approximation for

$$\int_2^4 f(x)\,dx$$

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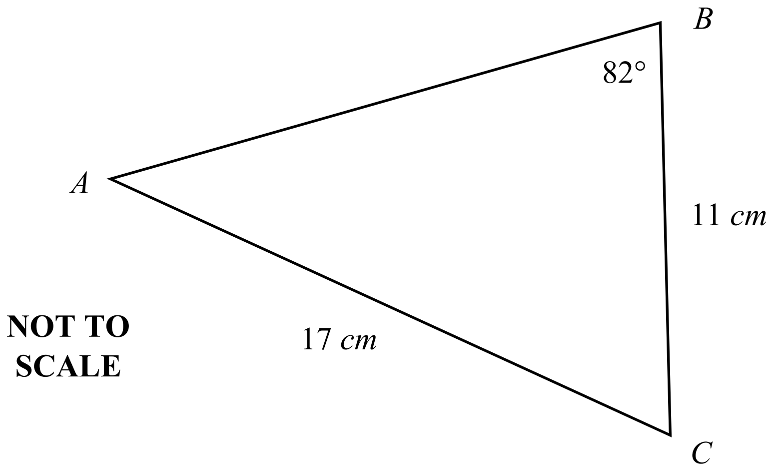
Question 14 (4 marks)

Triangle ABC is shown below



(a) Using the sine rule, show that $\angle A \approx 40°$. **2**

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(b) Hence, calculate the area of triangle ABC, correct to the nearest cm^2 **2**

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Question 15 (2 marks)

The stem-and-leaf plot shows the time taken, in minutes, for a group of students to travel to school each morning.

STEM	LEAF
0	7 9
1	2 3 3 8
2	0 4 4 5 9
3	5
4	1

(a) Find the median. **1**

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(b) Find the range. **1**

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Question 16 (2 marks)

Consider the arithmetic sequence 3, 7, 11, …

(a) Calculate the 10^{th} term. **1**

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(b) Calculate the sum of the first twenty terms. **1**

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Question 17 (3 marks)

Find the equation of the normal to the curve $y = x^2 + 3x$ at $x = 1$. **3**

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Question 18 (2 marks)

Solve $|x + 3| = 1$. **2**

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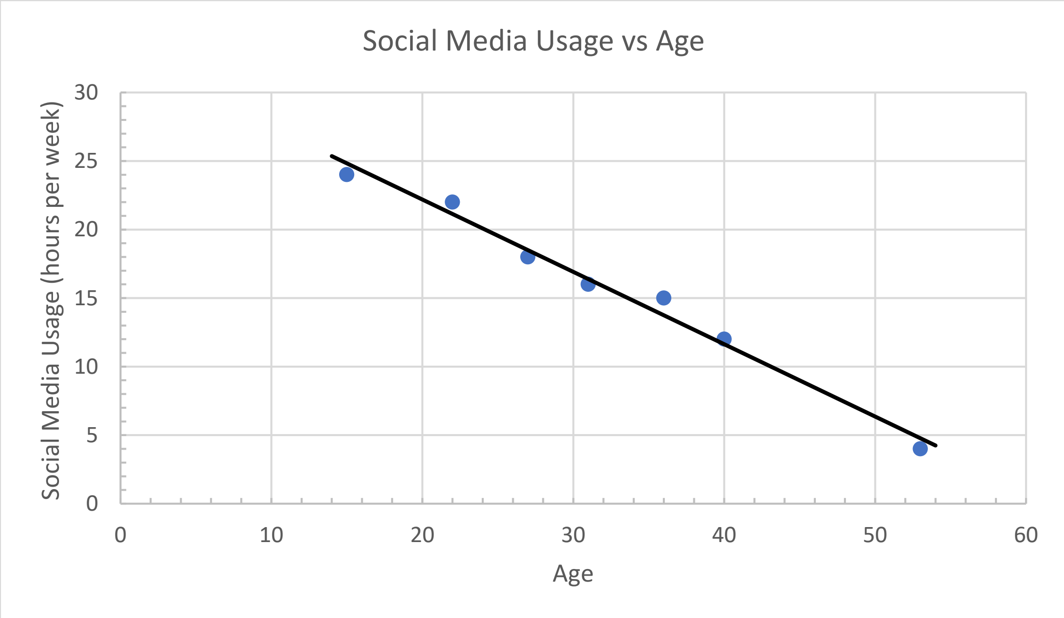
Question 19 (4 marks)

A group of people across a range of ages were surveyed about their social media usage.

The scatterplot below shows the hours spent each week on social media vs age.



The line of best fit for the data has also been added.

(a) The data is presented in a table below. **1**

Age	15	22	27	31	36	40	53
Hours per week	24	22	18	16	?	12	4

What is the missing value?

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(b) Calculate Pearson's correlation coefficient, correct to 4 significant figures. **1**

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Question 19 continues on page 13

Question 19 (continued)

(c) What type of correlation is suggested by this data? **1**

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(d) Determine the equation of the line of best fit, rounding coefficients to 2 decimal places. **1**

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Question 20 (4 marks)

The table below shows the probability distribution of a random variable X.

X	2	3	4	5
$P(X)$	0.4	0.1	0.27	0.23

(a) Calculate $E(X)$. **1**

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(b) Calculate $Var(X)$. **3**

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Question 21 (5 marks)

A class of 20 students was surveyed to find out how many movies they had watched in the past week. The data is presented in the frequency table below.

Number of movies (x)	**Frequency (f)**	$f \times x$
0	2	
1	7	
2	6	
3	3	
4	2	

(a) Complete the $f \times x$ column. **1**

(b) State the type of data that has been collected. **1**

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(c) State the modal number of movies watched. **1**

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(d) Find the mean number of movies watched. **1**

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(e) Find the population standard deviation, to three decimal places. **1**

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Question 22 (3 marks)

Show that the function $y = xe^x$ has a point of inflection at $x = -2$. **3**

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Question 23 (3 marks)

Given that $f'(x) = 9\cos 3x$ and $f(\pi) = 4$, find $f(x)$. **3**

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Question 24 (4 marks)

Sketch the graph of $y = \ln(x^2 + 1)$, showing any stationary points, given that there are points of inflection at $x = \pm 1$. **4**

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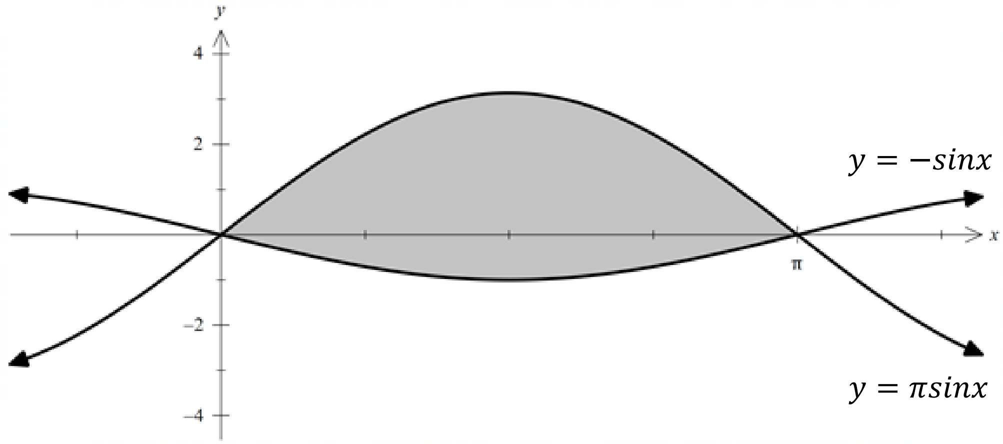
Question 25 (3 marks)

The graphs o $y = \pi \sin x$f and $y = -\sin x$ are shown below. **3**



Find the shaded area.

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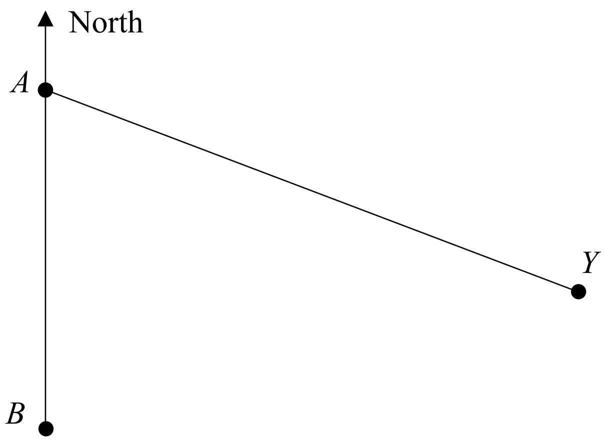
Questions 11-25 are worth 45 marks in total

Question 26 (5 marks)

A yacht leaves port A on a bearing of 120° and sails for three hours at an average speed of 15 km/h to its destination where it stops.

At the same time a speed boat also leaves from port A and travels due south to island B that is 30 km from the port.

NOT TO SCALE



(a) How far is the yacht from the port after 3 hours? **1**

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(b) Calculate the distance of the yacht from island B. **2**

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Question 26 continues on page 19

Question 26 (continued)

(c) After spending several hours at the island, the speed boat travels due north back to port A. How far south of the port will the speed boat be when it is directly west of the yacht? **2**

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Question 27 (2 marks)

The lifespans of a particle brand of battery are normally distributed with a mean of 90 hours and a standard deviation of 3 hours. **2**

The packaging claims that the batteries will have a lifespan of up to 96 hours.

If 160 batteries are tested, how many would be expected to exceed the claim on the packaging?

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Question 28 (4 marks)

A runner is training for a long-distance event.

The first week she runs 1.2 km.

The next week she runs 1.8 km.

The third week she runs 2.7 km and so on, adding on half of the previous week.

(a) How far does she run in the fourth week? **1**

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(b) The event she is training for is 45 km. In which week will she first exceed this distance? **3**

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Question 29 (3 marks)

The cost per hour, $\$C$, of running a train is dependent on the speed, v km/h, of the train according to the equation $C = \dfrac{v^2}{40} + \dfrac{10800}{v}$ **3**

Find the speed to minimise the cost per hour.

Question 30 (4 marks)

The volume of a pond decreases during a period of no rain.

The volume in the pond, V, after t days can be modelled by the equation

$$V = V_0 e^{-kt}$$

where V_0 is the initial volume of the pond.

(a) Given that it takes 10 days for the volume of the pond to halve, show that $k \approx 0.0693$. **3**

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(b) If there is 30 megalitres (ML) in the pond after 7 days, calculate the initial volume of the pond, to the nearest ML. **1**

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Question 31 (7 marks)

Henri has just opened a new bank account.

His monthly income of \$6000 is paid into the account at the beginning of each month.

The account earns interest of 3% p.a. with the interest calculated monthly.

After the interest has been calculated, Henri withdraws \$*M* at the end of each month to cover his living expenses.

Let A_n be the balance of the account after *n* months.

(a) Show that the balance of the account after 2 months is **2**

$$A_2 = 3000 \times (1.0025^2 + 1.0025) - M(1 + 1.0025)$$

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(b) Show that the amount in the account after *n* months is given by **3**

$$A_n = 3000 \times 401(1.0025^n - 1) - 400M(1.0025^n - 1)$$

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Question 31 continues on page 24

Question 31 (continued)

(c) Find the withdrawal amount, $\$M$, such that there will be \$15 000 in the account at the end of 2 years. **2**

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Question 32 (3 marks)

The population of a seal colony varies according to the equation

$$P = 100\sin\left(\frac{\pi t}{6}\right) + 300$$

where t is in months.

(a) Find the maximum and minimum populations of the colony. **1**

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(b) Find the number of months it takes for the population to vary from the maximum population to the minimum population. **2**

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Question 33 (7 marks)

A continuous random variable, X, is defined by the probability density function $P(X) = kx(3 - x)$ where $0 \leq x \leq 3$ and $k > 0$.

(a) Show that $k = \frac{2}{9}$ **2**

Question 33 continues on page 26

(b) Find the mode of X. **2**

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(c) Find the cumulative distribution function. **2**

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Question 33 continues on page 27

Question 33 (continued)

(d) Find the median value of X. **1**

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Question 34 (3 marks)

A particle is initially at the origin and moves along a straight line with its displacement given by

$$x = \frac{20\ln(1+t)}{(1+t)^2}$$

(a) Show that the velocity of the particle is given by **2**

$$v = \frac{20(1 - 2\ln(1+t))}{(1+t)^3}$$

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Question 34 continues on page 28

Question 34 (continued)

(b) Describe what happens to the displacement and velocity eventually. **1**

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Question 35 (7 marks)

A 'random walk' is generated by having a person, initially at the origin, flip a coin.

If the coin shows heads, the person takes one step forwards.

If the coin shows tails, the person takes one step backwards.

Both outcomes are equally likely.

After n steps, the position of the person will be some integer number of steps from the origin.

This final position of the person after n steps can be represented by the discrete random variable $X(n)$.

(a) Explain why $X(n+1) = X(n) \pm 1$ **1**

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Question 35 continues on page 29

(b) Show that the mean value of $[X(n+1)]^2$ is $[X(n)]^2 + 1$ **2**

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Let $|X(n)|$ represent the distance that an $n-$step walk is from the origin.

(c) What is the value of $|X(1)|$? **1**

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Question 35 continues on page 30

Question 35 (continued)

(d) Given that $[X(1)]^2 = 1$, show that $E[|X(n)|] = \sqrt{n}$ **3**

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End of Paper

Mathematics Advanced

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface Area

$$A = 2\pi r^2 + 2\pi r h$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

and $$\alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and Series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, \ r \neq 1$$

$$S = \frac{a}{1-r}, \quad |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

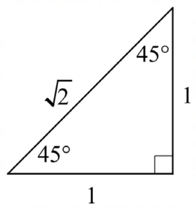
$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab\sin C$$



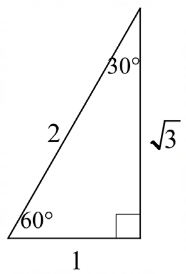
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$\sec A = \dfrac{1}{\cos A}, \ \cos A \neq 0$

$\operatorname{cosec} A = \dfrac{1}{\sin A}, \ \sin A \neq 0$

$\cot A = \dfrac{\cos A}{\sin A}, \ \sin A \neq 0$

$\cos^2 x + \sin^2 x = 1$

Compound angles

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

If $t = \tan\dfrac{A}{2}$ then $\sin A = \dfrac{2t}{1+t^2}$

$\cos A = \dfrac{1-t^2}{1+t^2}$

$\tan A = \dfrac{2t}{1-t^2}$

$\cos A \cos B = \dfrac{1}{2}\left[\cos(A-B) + \cos(A+B)\right]$

$\sin A \sin B = \dfrac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$ $\sin A \cos B = \dfrac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$

$\cos A \sin B = \dfrac{1}{2}\left[\sin(A+B) - \sin(A-B)\right]$

$\sin^2 nx = \dfrac{1}{2}(1 - \cos 2nx)$

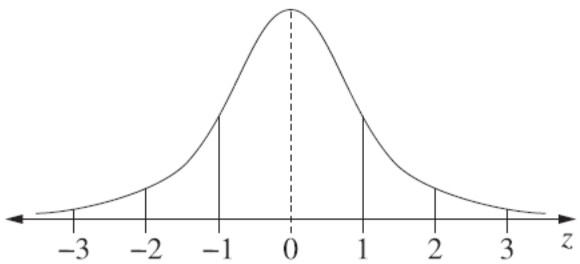
$\cos^2 nx = \dfrac{1}{2}(1 + \cos 2nx)$

Statistical Analysis

$z = \dfrac{x-\mu}{\sigma}$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal Distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X)=\mu$$

$$\operatorname{Var}(X)=E\left[(X-\mu)^2\right]=E\left(X^2\right)-\mu^2$$

Probability

$$P(A\cap B)=P(A)P(B)$$

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)\quad P(A|B)=\frac{P(A\cap B)}{P(B)},\quad P(B)\neq 0$$

Continuous random variables

$$P(X\leq r)=\int_a^r f(x)\,dx$$

$$P(a<X<b)=\int_a^b f(x)\,dx$$

Binomial distribution

$$P(X=r)={}^nC_r\,p^r(1-p)^{n-r}$$

$$X\sim\operatorname{Bin}(n,p)$$

$$\Rightarrow\quad P(X=x)$$

$$=\binom{n}{x}p^x(1-p)^{n-x},\ x=0,1,\ldots,n$$

$$E(X)=np$$

$$\operatorname{Var}(X)=np(1-p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = nf'(x)\left[f(x)\right]^{n-1}$
$y = uv$	$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x)\cos f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-\left[f(x)\right]^2}}$
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-\left[f(x)\right]^2}}$

$y = \tan^{-1} f(x)$ $\qquad$ $\dfrac{dy}{dx} = \dfrac{f'(x)}{1+\left[f(x)\right]^2}$

Integral Calculus

$$\int f'(x)\left[f(x)\right]^n dx = \frac{1}{n+1}\left[f(x)\right]^{n+1} + c$$

where $n \neq 1$

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln\left|f(x)\right| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2-\left[f(x)\right]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2+\left[f(x)\right]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)\,dx$$
$$\approx \frac{b-a}{2n}\left\{f(a)+f(b)+2\left[f(x_1)+\ldots+f(x_{n-1})\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \ldots + \binom{n}{r}x^{n-r}a^r + \ldots + a^n$$

Vectors

$$|\underset{\sim}{u}| = |x\underset{\sim}{i} + y\underset{\sim}{j}| = \sqrt{x^2+y^2}$$

$$\underset{\sim}{u}\cdot\underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$
$$= re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i\sin n\theta)$$
$$= r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt+\alpha) + c$$

$x = a\sin\left(nt + \alpha\right) + c$

$\ddot{x} = -n^2\left(x - c\right)$

Student Number:…………………………

MATHEMATICS ADVANCED – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

Question	**1**	A ◯	B ◯	C ◯	D ◯
	2	A ◯	B ◯	C ◯	D ◯
	3	A ◯	B ◯	C ◯	D ◯
	4	A ◯	B ◯	C ◯	D ◯
	5	A ◯	B ◯	C ◯	D ◯
	6	A ◯	B ◯	C ◯	D ◯
	7	A ◯	B ◯	C ◯	D ◯
	8	A ◯	B ◯	C ◯	D ◯
	9	A ◯	B ◯	C ◯	D ◯
	10	A ◯	B ◯	C ◯	D ◯

2022 Higher School Certificate

Mathematics Advanced
Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	B
3	C
4	B
5	C
6	D
7	A
8	D
9	B
10	A

Section II

Question 11

Criteria	Mark
• Correct answer	1

Sample answer

$$\frac{d}{dx}(x+4)^7 = 7(x+4)^6$$

Question 12

Criteria	Marks
• Correct answer	2
• Correct primitive with incorrect substitution or calculation error	1

Sample answer

$$\int_0^1 e^{5x}\,dx = \left[\frac{1}{5}e^{5x}\right]_0^1$$

$$= \left(\frac{e^5}{5} - \frac{e^0}{5}\right)$$

$$= \frac{e^5 - 1}{5}$$

Question 13

Criteria	Marks
• Correct answer	3
• Correct use of trapezoidal rule with calculation error	2
• Use only one application of trapezoidal rule or incorrect *n* value	1

Sample answer

$$\int_2^4 f(x)\,dx \approx \frac{1}{2}[6 + 4 + 2 \times 5]$$

$$= 10$$

Question 14a

Criteria	Marks
• Correctly shows value	2
• Attempts to use sine rule	1

Sample answer

$$\frac{\sin A}{11} = \frac{\sin 82}{17}$$

$$\sin A = \frac{11 \sin 82}{17}$$

$$\therefore A = 39.8486 \ldots \approx 40°$$

Question 14b

Criteria	Marks
• Correct answer	2
• Finds value of ∠C	1

Sample answer

$$C = 180 - 82 - 40 = 58°$$

$$Area = \frac{1}{2} \times 11 \times 17 \times \sin 58$$

$$= 79.292 \ldots \approx 79 \text{ cm}^2$$

Question 15a

Criteria	Mark
• Correct answer	1

Sample answer

Median = 20

Question 15b

Criteria	Mark
• Correct answer	1

Sample answer

Range = 41 – 7 = 34

Question 16a

Criteria	Mark
• Correct answer	1

Sample answer

$T_{10} = 3 + (10 - 1) \times 4 = 39$

Question 16b

Criteria	Mark
• Correct answer	1

Sample answer

$$S_{10} = \frac{20}{2}(2 \times 3 + (20 - 1) \times 4) = 820$$

Question 17

Criteria	Marks
• Correct answer	3
• Finds gradient of normal	2
• Finds correct derivative	1

Sample answer

$y' = 2x + 3$

At $x = 1, y = 4$ and $y' = 5$

$$\therefore m_T = 5 \rightarrow m_N = -\frac{1}{5}$$

Equation of normal:

$$y - 4 = -\frac{1}{5}(x - 1)$$

$\therefore x + 5y - 21 = 0$

Question 18

Criteria	Marks
• Correct solutions	2
• Only provides one correct solution	1

Sample answer

$x + 3 = 1$ or $x + 3 = -1$

$\therefore x = -2$ or $x = -4$

Question 19a

Criteria	Mark
• Correct answer	1

Sample answer

Missing value = 15

Question 19b

Criteria	Mark
• Correct answer	1

Sample answer

r = -0.9922

Question 19c

Criteria	Mark
• Correct answer	1

Sample answer

Strong, negative correlation.

Question 19d

Criteria	Mark
• Correct answer	1

Sample answer

$y = 32.7 - 0.53x$

Question 20a

Criteria	Mark
• Correct answer	1

Sample answer

$E(X) = 2 \times 0.4 + 3 \times 0.1 + 4 \times 0.27 + 5 \times 0.23 = 3.33$

Question 20b

Criteria	Marks
• Correct answer	3
• Progress towards correct answer with calculation error	2
• Attempts to use either form of variance formula	1

Sample answer

$Var(X) = 2^2 \times 0.4 + 3^2 \times 0.1 + 4^2 \times 0.27 + 5^2 \times 0.23 - (3.33)^2 = 1.4811$

Question 21a

Criteria	Mark
• Correct answer	1

Sample answer

Number of movies (x)	**Frequency (f)**	$f \times x$
0	2	**0**
1	7	**7**
2	6	**12**
3	3	**9**
4	2	**6**

Question 21b

Criteria	Mark
• Correct answer	1

Sample answer

Discrete numerical.

Question 21c

Criteria	Mark
• Correct answer	1

Sample answer

Mode = 1

Question 21d

Criteria	Mark
• Correct answer	1

Sample answer

Mean = 1.8

Question 21e

Criteria	Mark
• Correct answer	1

Sample answer

Standard Deviation = 1.122

Question 22

Criteria	Marks
• Correctly shows inflection	3
• Shows $y'' = 0$ but doesn't show concavity changes	2
• Finds y'' correctly	1

Sample answer

$y = xe^x$

$y' = e^x + xe^x$

$y'' = xe^x + 2e^x$

When $x = -2,\ y'' = -2e^{-2} + 2e^{-2} = 0$

Check concavity:

x	-3	-2	-1
y''	$-e^{-3} < 0$	0	$e^{-1} > 0$

Concavity changes.

$\therefore$ there is an inflection at $x = -2$

Question 23

Criteria	Marks
• Correct solution	3
• Incorrect calculation of constant	2
• Correctly finds primitive	1

Sample answer

$f(x) = 3\sin 3x + C$

$3\sin 3\pi + C = 4$

$0 + C = 4$

$\therefore C = 4$

$\therefore f(x) = 3\sin 3x + 4$

Question 24

Criteria	Marks
• Correct graph showing stationary point	4
• Correct stationary point with incorrect shape as $x \to \pm\infty$	3
• Finds stationary point correctly	2
• Finds y' correctly	1

Sample answer

$y = \ln(x^2 + 1)$

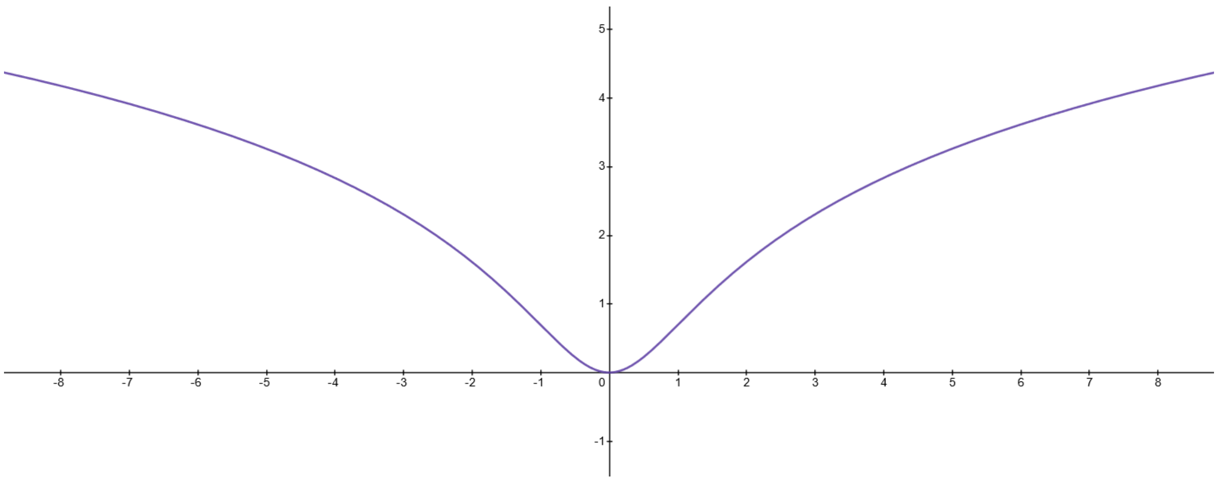
$$y' = \frac{2x}{x^2 + 1}$$

$y' = 0 \to x = 0$

when $x = 0, y = 0$

x	-1	0	1
y'	$-1 < 0$	0	$1 > 0$

$(0,0)$ is a minimum turning point.



Question 25

Criteria	Marks
• Correct solution	3
• Finds correct primitive	2
• Sets up correct integral	1

Sample answer

$$Area = \int_0^{\pi} (\pi \sin x - -\sin x) dx$$

$$= (\pi + 1) \int_0^{\pi} \sin x \, dx$$

$$= (\pi + 1)[-\cos x]_0^{\pi}$$

$$= (\pi + 1)[-\cos \pi + \cos 0]$$

$= 2(\pi + 1)$ units^2

Question 26a

Criteria	Mark
• Correct answer	1

Sample answer

Distance = 15 × 3 = 45 km

Question 26b

Criteria	Marks
• Correct solution	2
• Correct substitutions into cosine rule with error in calculation	1

Sample answer

$$d^2 = 45^2 + 30^2 - 2 \times 45 \times 30 \times \cos 60°$$

$$d^2 = 1575$$

$d = \sqrt{1575} = 39.686 \ldots \approx 40$ km

Question 26c

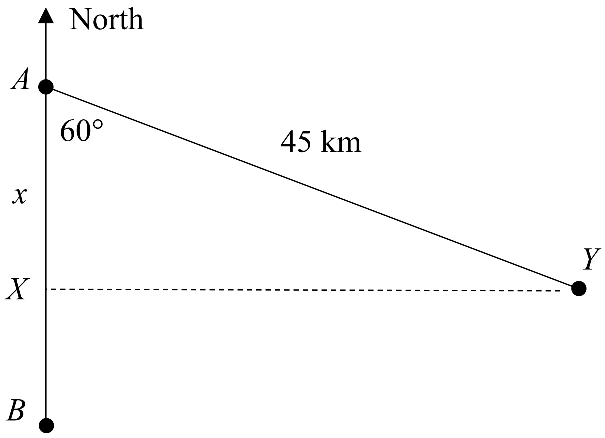
Criteria	Marks
• Correct solution	2
• Attempts to use right-angled trigonometry to solve	1

Sample answer

$\cos 60 = \dfrac{x}{45}$

$x = 45 \cos 60$

$\therefore x = 22.5$ km



Question 27

Criteria	Marks
• Correct solution	2
• Correct percentage from z-score	1

Sample answer

$z = \dfrac{96 - 90}{3} = 2$

For normal distribution, 2.5% > z-score of 2.

$160 \times 2.5\% = 4$

4 batteries should exceed 96 hours.

Question 28a

Criteria	Mark
• Correct answer	1

Sample answer

$T_4 = 1.2 \times 1.5^{4-1} = 4.05$ km

Question 28b

Criteria	Marks
• Correct answer	3
• Correct solution to exponential, but incorrectly interprets result	2
• Attempts to solve correct exponential equation	1

Sample answer

$45 = 1.2 \times 1.5^{n-1}$

$1.5^{n-1} = 37.5$

$\ln 1.5^{n-1} = \ln 37.5$

$$n - 1 = \frac{\ln 37.5}{\ln 1.5}$$

$n - 1 = 8.9387 \ldots$

$n = 9.9387 \ldots$

She will exceed 45 km in the tenth week.

Question 29

Criteria	Marks
• Correct answer	3
• Correctly finds *v*, but doesn't show it is a minimum point	2
• Correct derivative	1

Sample answer

$$C' = \frac{v}{20} - \frac{10800}{v^2}$$

$C' = 0$ for min/max

$$\frac{v}{20} - \frac{10800}{v^2} = 0$$

$$v^3 - 216000 = 0$$

$$v = 60$$

$C'' = \frac{1}{20} + \frac{21600}{v^3} > 0$ for all $v > 0$

Therefore minimum point.

60 km/h gives minimum cost.

Question 30a

Criteria	Marks
• Correct solution	3
• Rewrites exponential as a logarithm	2
• Uses $V = 0.5V_0$	1

Sample answer

$0.5V_0 = V_0 e^{-10k}$

$e^{-10k} = 0.5$

$-10k = \ln 0.5$

$k = \dfrac{\ln 0.5}{-10} = 0.0693147 \ldots \approx 0.0693$

Question 30b

Criteria	Mark
• Correct solution	1

Sample answer

$30 = V_0 e^{-0.0693 \times 7}$

$V_0 = \frac{30}{e^{-0.4851}} = 48.730 \ldots \approx 49$ ML

Question 31a

Criteria	Marks
• Correct expression for A_2 derived from A_1	2
• Correct expression for A_1	1

Sample answer

$A_1 = 3000 \times 1.0025 - M$

$A_2 = (A_1 + 3000) \times 1.0025 - M$

$A_2 = (3000 \times 1.0025 - M + 3000) \times 1.0025 - M$

$A_2 = 3000 \times (1.0025^2 + 1.0025) - M(1 + 1.0025)$

Question 31b

Criteria	Marks
• Uses sum of GP to generate required expression	3
• Generalises for A_n	2
• Shows A_3	1

Sample answer

$A_3 = (A_2 + 3000) \times 1.0025 - M$

$= 3000 \times (1.0025^3 + 1.0025^2 + 1.0025) - M(1 + 1.0025 + 1.0025^2)$

$A_n = 3000 \times (1.0025^n + \cdots + 1.0025^2 + 1.0025) - M(1 + 1.0025 + \ldots + 1.0025^{n-1})$

$1 + 1.0025 + \ldots + 1.0025^{n-1}$ is a GP with $a = 1, r = 1.0025$ and *n* terms

$(1.0025^n + \cdots + 1.0025^2 + 1.0025)$ is a GP with $a = 1.0025, r = 1.0025$ and *n* terms

$$\therefore A_n = 3000\left(\frac{1.0025(1.0025^n - 1)}{1.0025 - 1}\right) - M\left(\frac{1(1.0025^n - 1)}{1.0025 - 1}\right)$$

$\therefore A_n = 3000 \times 401(1.0025^n - 1) - 400M(1.0025^n - 1)$

Question 31c

Criteria	Marks
• Finds correct value	2
• Correct substitution	1

Sample answer

$15000 = 3000 \times 401(1.0025^{24} - 1) - 400M(1.0025^{24} - 1)$

$$M = \frac{3000 \times 401(1.0025^{24} - 1) - 15000}{400(1.0025^{24} - 1)}$$

$M = \$2400.28$

Question 32a

Criteria	Mark
• Correct answer	1

Sample answer

max = 400, min = 200

Question 32b

Criteria	Marks
• Correct answer	2
• Correct period	1

Sample answer

$$P = \frac{2\pi}{\frac{\pi}{6}} = 12$$

6 months between maximum and minimum.

Question 33a

Criteria	Marks
• Correct answer	2
• Correct integral	1

Sample answer

$$\int_0^3 kx(3-x)\,dx = 1$$

$$k\left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = 1$$

$$k\left(\frac{27}{2} - 9\right) = 1$$

$$\therefore k = \frac{2}{9}$$

Question 33b

Criteria	Marks
• Correct answer	2
• Attempts to find vertex of parabola	1

Sample answer

The mode occurs at vertex of the parabola.

x-intercepts are 0 and 3.

Vertex is halfway between, $x = 1.5$

Therefore, mode is 1.5

Question 33c

Criteria	Marks
• Correct answer	2
• Correct integral	1

Sample answer

$$CDF = \int_0^x \frac{2}{9} t(3 - t)\, dt$$

$$= \frac{2}{9}\left[\frac{3t^2}{2} - \frac{t^3}{3}\right]_0^x$$

$$= \frac{2}{9}\left(\frac{3x^2}{2} - \frac{x^3}{3}\right) - 0$$

$$\therefore CDF = \frac{2}{9}\left(\frac{3x^2}{2} - \frac{x^3}{3}\right)$$

Question 33d

Criteria	Mark
• Correct answer	1

Sample answer

By the symmetry of the function, the median coincides with the vertex of the parabola.

Hence median is 1.5

Question 34a

Criteria	Marks
• Correctly simplifies to required form	2
• Correctly applies quotient rule	1

Sample answer

$$v = \frac{\frac{20}{1+t} \times (1+t)^2 - 2(1+t) \times 20\ln(1+t)}{(1+t)^4}$$

$$v = \frac{20(1+t) - 40(1+t)\ln(1+t)}{(1+t)^4}$$

$$v = \frac{20 - 40\ln(1+t)}{(1+t)^3}$$

$$\therefore v = \frac{20(1 - 2\ln(1+t))}{(1+t)^3}$$

Question 34b

Criteria	Mark
• Correct answer for displacement or velocity	1

Sample answer

As $t \to \infty, x \to 0^+$ and $v \to 0^-$

Thus, the particle approaches the origin while constantly slowing.

Question 35a

Criteria	Mark
• Correct explanation	1

Sample answer

From any position $X(n)$ the person can only take 1 step forward or 1 step backwards.

Hence $X(n+1)$ can only vary from $X(n)$ by ± 1

Question 35b

Criteria	Marks
• Correctly shows expression.	2
• Correct expressions for $[X(n+1)]^2$	1

Sample answer

$X(n+1) = X(n) + 1$ or $X(n+1) = X(n) - 1$

$[X(n+1)]^2 = [X(n)+1]^2$ or $[X(n+1)]^2 = [X(n)-1]^2$

$= [X(n)]^2 + 2X(n) + 1$ $\quad$ $= [X(n)]^2 - 2X(n) + 1$

Hence, the mean value is:

$$\frac{[X(n)]^2 + 2X(n) + 1 + [X(n)]^2 - 2X(n) + 1}{2}$$

$$= \frac{2[X(n)]^2 + 2}{2}$$

$$= [X(n)]^2 + 1$$

Question 35c

Criteria	Mark
• Correct answer	1

Sample answer

$|X(1)| = 1$

Question 35d

Criteria	Marks
• Uses $\lvert a\rvert = \sqrt{a^2}$ to generate required expression	3
• Establishes the result $E\{[X(n)]^2\} = n$ using (b)	2
• States $E\{[X(1)]^2\} = 1$	1

Sample answer

$[X(1)]^2 = 1$

$\therefore E\{[X(1)]^2\} = 1$

From (b); $E\{[X(n+1)]^2\} = E\{[X(n)]^2\} + 1$

$\therefore E\{[X(2)]^2\} = E\{[X(1)]^2\} + 1 = 1 + 1 = 2$

And

$E\{[X(3)]^2\} = E\{[X(2)]^2\} + 1 = 2 + 1 = 3$

And so on.

$\therefore E\{[X(n)]^2\} = n$

Also, $\lvert X(n)\rvert = \sqrt{[X(n)]^2}$

$\therefore E[\lvert X(n)\rvert] = \sqrt{n}$

2022 Higher School Certificate
Trial Examination
Mathematics
Mapping Grid

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Pareto chart	MA-S2.1*
2	1	Exact trigonometry	MA-T2
3	1	Mean	MA-S2.1*
4	1	Simple probability	MA-S1.1*
5	1	Integration	MA-C4.2
6	1	Trigonometric graph	MA-T3
7	1	Discrete probability	MA-S1.2
8	1	Transformation of graphs	MA-F2
9	1	Gradient of tangent	MA-C1.4
10	1	Odd and Even functions	MA-F1.2

Section II

Question	Marks	Content	Syllabus Outcomes
11	1	Differentiation	MA-C2
12	2	Definite integral	MA-C4.2
13	3	Trapezoidal rule	MA-C4.2*
14	4	Non-right-angled trigonometry	MA-T1.1*
15	2	Data – median and range	MA-S2.1*
16	2	Arithmetic series	MA-M1.2
17	3	Equation of normal	MA-C1.4
18	2	Absolute value equation	MA-F1.4
19	4	Bivariate data	MA-S2.2*
20	4	Discrete Probability	MA-S1.2
21	5	Data - mean and s.d.	MA-S2.1*
22	3	Point of inflection	MA-C3
23	3	Primitive function	MA-C4.1
24	4	Curve sketching	MA-C3
25	3	Area between curves	MA-C4.2
26	5	Trigonometry	MA-T1.1*
27	2	Normal distribution	MA-S3.2*
28	4	Geometric series	MA-M1.3
29	3	Max/min problem	MA-C3.2
30	4	Exponential decay	MA-E1.4
31	7	Financial application of series	MA-M1.4
32	3	Application of trig graphs	MA-T3
33	7	Continuous probability	MA-S3.1
34	3	Motion	MA-C4.1
35	7	Discrete Random Variable	MA-S1.2